OPERATIONS · AGRONOMY · ENOLOGY · HOSPITALITY · REGISTER

Tenuta Baiamonte System Manual

Operations, agronomy, enology, hospitality, register, security, and recovery.

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CONTENTS

System at a glance

This manual follows the live dashboard from everyday use through administration, recovery, and release verification.

This manual describes what the Baiamonte system does, how staff use it, how its calculations and predictions work, how data moves between services, and how to recognize and recover from problems. It is written for owners, vineyard staff, cellar staff, accountants, agronomists, kiosk users, and technical administrators.

OPERATING REFERENCE

1. What the system is

Tenuta Baiamonte Vineyard Operations is a private estate-management system hosted as a Home Assistant add-on. It combines vineyard work, harvest, cellar, laboratory, weather, treatment, labor, payment, olive-oil, hospitality, messaging, alert, map, and forecasting functions in one controlled interface.

The core rule is simple:

> MariaDB is the operational authority. Historical workbooks, messages, notes, emails, and laboratory documents are evidence sources, not live databases.

Imported evidence is reviewed, normalized, and written to MariaDB. The dashboards, predictions, labels, payment views, and reports read from MariaDB. The old workbooks are not consulted during normal operation.

System goals

- Give each role a useful, limited view of the estate.
- Preserve historical records by year and vintage.
- Distinguish facts, estimates, forecasts, and unresolved questions.
- Keep treatment and cellar decisions under human approval.
- Make labor approval and payment auditable.
- Use free, credential-free environmental sources where practical.
- Continue operating through temporary source outages using stored last-good data.
- Keep secrets in protected Home Assistant add-on configuration.

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2. System map

Component	Purpose	Authority / access
Home Assistant	Users, devices, dashboards, cameras, weather entities, Supervisor	Home Assistant accounts
Vineyard Operations	Main web interface and business logic	Home Assistant Ingress
Hospitality	Private tastings, dinners, guest readiness, deposits, and communication history	Hospitality Manager or Administrator
Register	Tablet-oriented sales, sellable inventory, receipts, and monthly transaction ledger	Register, Cashier, Hospitality Manager, or Administrator
MariaDB	Sole operational database	Add-on network only
Baiamonte MCP	Controlled Codex and automation bridge	Bearer token; local/VPN
Vineyard TV	Read-only rotating operational display	Viewer profile
iPad dashboard	Larger touch operations dashboard	`ipad` viewer profile
Cellar labels	Tank identity, live readings, charts, legal identity	Enrolled kiosk URLs
Gmail intake	Email and attachment review queue	Protected mailbox credentials
WhatsApp	Approved inbound/outbound operational messaging	Meta or system bridge credentials
Fatture in Cloud	Read-only accounting mirror	Protected read-only token
Public harvest feed	Approved public harvest dates only	Optional publication token

Network addresses

- Home Assistant: `http://192.168.0.10:8123`
- Vineyard Operations direct service: `http://192.168.0.10:8101`
- Baiamonte MCP: `http://192.168.0.10:8100/mcp`

Use Home Assistant Ingress for normal staff access. Direct local addresses are intended for service integration, diagnosis, enrolled displays, and VPN use.

OPERATING REFERENCE

3. Users and permissions

Administrators

Administrators can manage system configuration, users, process controls, messaging, labor approval, payroll, documentation, and diagnostics. Administrative access should remain limited to trusted owners or system managers.

Operations users

Operations users can manage vineyard work, harvest, cellar records, laboratory review, olives, treatments, issues, and operational intake. They do not automatically receive full finance or system-configuration access.

Finance users

Finance users can view the read-only Fatture in Cloud mirror, invoices, payments, expenses, and financial reports. Finance information is intentionally excluded from shared TV and kiosk feeds.

Worker users

Worker accounts use a simplified personal portal to clock in and out, enter services or expenses, attach evidence, see review status, and view approved history. A worker cannot approve or pay their own record.

Viewer users

The `display`, `tv`, and `ipad` profiles are read-only. They do not receive finance data or administrative controls.

Hospitality Managers

Hospitality Managers can manage private tastings and dinners, guest requirements, quotes, deposits, confirmations, arrival, and completion. They do not automatically receive vineyard write, finance, payroll, or system-administration access.

Register and Cashier users

Register and Cashier users can sell the read-only Fatture in Cloud product catalog, local hospitality packages, and authorized manual items. They can apply a line discount, choose EUR or USD tender, select the Italian or US PayPal Business account, print a non-fiscal operational receipt, and export the monthly local ledger. They do not automatically receive finance, payroll, vineyard-write, or administration access.

Home Assistant People and Users are authoritative for identity, display name, username, profile picture, and presence. Vineyard Operations is authoritative for estate access level, operational role, approval authority, and hospitality permissions. An administrator should assign access and roles in **Admin -> People** rather than creating a second identity.

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4. First use and navigation

1. Sign in to Home Assistant with the appropriate account.
2. Open **Vineyard Operations** from the sidebar.
3. Choose **Operations**, **Agronomy**, **Enology**, **Hospitality**, **Register**, or **Admin** from the top workspace switch. Only authorized workspaces are shown.
4. Choose the working year. Year selection changes the harvest, laboratory, weather comparison, labor, treatment, olive, and historical context.
5. Use the main sections for daily work; use Administration only for configuration or audit tasks.
6. Treat yellow or amber items as uncertain or awaiting review. Treat red items as active exceptions, not automatically as failed equipment.

Changing the working year keeps the current workspace and page open. For example, changing vintage while viewing Enology -> Laboratory refreshes Laboratory for that vintage; it does not move the user to Operations. A page can explicitly show the latest earlier vintage when the selected year has no measurements, but it must label that fallback and must not present it as current-year evidence.

Main work areas

- **Today:** current work, recent work, estate status, weather, and alerts.
- **Vines / Harvest:** variety status, harvest plans, actual picks, predictions, and vintage history.
- **Treatments:** forecasts, planned work, approvals, completed applications, and safety evidence.
- **Cellar:** wine lots, vessels, readings, operations, labels, and tank assignments.
- **Labor / Payroll:** work records, approval, contractor invoices, deposits, and payments.

- **Olives:** harvest and oil history, conversion efficiency, cost assumptions, and YoY charts.
- **Laboratory:** reports, vintage assignment, analytes, trends, review, and decision support.
- **Atlas:** cadastral parcels, vineyard blocks, terraces, nursery data, and map geometry.
- **Intelligence:** weather, GDD, seasonal comparisons, prediction evidence, disease pressure, and outlooks.
- **Hospitality:** private experiences, packages, bookings, guest readiness, deposits, and communication history.
- **Register:** tablet sales, sellable inventory, receipts, transaction exports, PayPal account selection, and register settings.
- **Inbox / Messaging:** Gmail and WhatsApp intake, review, contacts, delivery, and replies.
- **Alerts:** current operational alerts and delivery settings.
- **Administration:** process health, users, integrations, logs, documentation, and update controls.

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5. Data authority and evidence rules

Every value should be understood as one of four types:

1. **Fact:** a confirmed measurement or completed event.
2. **Estimate:** a calculated value based on stated inputs.
3. **Forecast:** a future projection with confidence and evidence.
4. **Unresolved:** missing, conflicting, or awaiting human verification.

Blank database values mean **unknown**, not zero. The system should never silently convert a missing quantity into zero.

Evidence flow

1. A message, email, note, document, sensor reading, or manual entry arrives.
2. The item is stored with its original evidence.
3. Automated extraction may propose structured records.
4. A person reviews material changes.
5. Approved facts are saved to MariaDB with an audit trail.
6. Dependent dashboards and models are refreshed.

Vintage assignment

Laboratory and cellar reports are assigned to the correct vintage rather than merely the document year. Reports that cross calendar years stay with the wine vintage they describe. Explicit vintage labels and linked wine lots take precedence; inferred assignments remain marked with confidence and evidence.

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6. Intelligence pipelines and decision workflows

The intelligence layer is a collection of evidence pipelines, not one autonomous decision maker. Each pipeline receives evidence, creates a traceable calculation or proposal, identifies missing information, and hands the result to the person or record that has authority. The system may use a provisional result while review is pending only where the workflow explicitly allows it; it never converts a prediction into a completed event.

Common intelligence contract

Every intelligence result carries the same operating context:

- **Source and timestamp:** where the evidence came from and when it was observed.
- **Scope:** estate, zone, block, parcel, variety, lot, tank, reservation, person, or financial document.
- **Evidence state:** original attachment or message, extracted facts, conflicts, and missing inputs.
- **Calculation state:** model/rule version, result, range, confidence, and explanatory factors.
- **Authority state:** draft, proposed, approved, completed, rejected, superseded, or unresolved.
- **Audit state:** who reviewed or changed the result and what downstream records were refreshed.

An approved fact remains separate from the next system proposal. Recalculation can create a new proposal, but it does not silently overwrite an agronomist-approved loss, an approved treatment, a confirmed harvest plan, a reviewed laboratory result, a sent guest message, or an accounting document.

Master evidence router

```
New evidence
  |
  +-- hail / frost / wind / physical injury
  |      -> damage event chain -> yield impact proposal
  |
  +-- mildew / mold / pest / disease symptom
  |      -> disease pressure -> treatment review
  |
  +-- phenology / berry maturity / grape laboratory result
  |      -> harvest readiness -> harvest forecast
  |
  +-- wine or must laboratory result
  |      -> enology review -> lot and cellar history
  |
  +-- weather observation or forecast
  |      -> disease, treatment timing, harvest and alerts
  |
  +-- email / message / uploaded document
         -> extraction review -> approved domain record
```

The submitter chooses the observed condition and growth stage from controlled lists. The router preserves the original evidence and can attach a new observation to an existing issue chain, such as the estate-wide 2026 hail event, instead of creating a duplicate event.

Damage assessment and forecast-impact pipeline

Damage intelligence estimates crop impact while keeping the system estimate independent from the agronomist's final decision. Estate-wide means 100% geographic coverage of the estate; it does not mean 100% yield loss.

```
Event or follow-up report
  -> identify event chain and geographic scope
  -> evaluate structured scouting and optional photos
  -> calculate AI loss %, range and confidence
```

```
  -> compare with earlier reports and approved final
  -> create a new proposal for the chain
  -> agronomist reviews and approves or edits
  -> approved chain final adjusts the harvest forecast
```

Each report card shows the result produced by that report. The chain summary shows the current system proposal, the latest approved agronomist value, and the percentage currently used by forecasting. Follow-up reports refine the chronological assessment; they do not rewrite the historical result of an earlier card. Until an updated proposal is approved, the last approved chain final remains authoritative. If no agronomist value exists yet, the clearly labelled provisional AI value may feed planning with its confidence range.

Disease pressure and treatment pipeline

Weather, disease pressure, field scouting, crop, growth stage, treatment history, product evidence, inventory, sprayer configuration, and harvest clearance operate as one decision chain. Vineyard and olive programs are evaluated separately.

```
Weather + field evidence + crop stage + treatment cadence
  -> calculate disease/stress pressure by target
  -> decide whether review is needed
  -> screen crop-authorized products and support products
  -> check season, target, rate, PHI/REI and compatibility
  -> calculate homogeneous mixture and sprayer batches
  -> compare required quantity with the inventory ledger
  -> find a defensible application window
  -> agronomist approves product, date and instructions
  -> operator records the completed application
```

The output can contain a primary disease-control product plus justified nutrition, wound support, wetting, resistance-management, or other support products. A product is included only when its current label/formulation evidence supports the crop, target, rate, unit, timing, and combination. Separate passes are shown when same-tank compatibility is not verified. Every prepared tank is treated as a homogeneous mixture under agitation.

The treatment simulator replays the same evidence and rules for a hypothetical or historical date but saves nothing and authorizes nothing. It shows the proposed products, amounts, batches, timing, evidence gaps, on-hand stock, shortage, and total required inventory. A negative stock ledger is permitted when completed use precedes a delayed invoice; the shortage remains an operational issue until a receipt nets it or an authorized person resolves it as not needed for the rest of the season.

For Baiamonte vineyard work, the default operating picture is one pass over the complete selected vineyard area using 400 L total carrier, prepared as two 200 L fills with the same approved recipe. The simulator adds only products supported by the current target, field evidence, crop stage, historical Agronomist pattern, inventory and label rules. It does not add a nutrition, support or disease-control product merely because the product exists in inventory. When exact same-tank compatibility is not recorded, the program remains blocked or separates the pass for Agronomist review.

The disease model recalculates pressure from GW2000 temperature, humidity, rainfall, leaf wetness when available, soil moisture when available, wind, solar conditions, phenology, maturity evidence, localized scouting and recent treatment context. Weather is the principal rationale for timing, while field evidence determines whether pressure is actually actionable. The model also creates disease-onset forecasts, block-specific calibration, spray-window evidence, expected product duration, retreatment cadence and resistance-rotation review. These learned outputs remain provisional until sufficient reviewed outcomes exist.

Agronomist review records the accepted risk level or corrected score and the reason. Paired pre/post-treatment scouting links the last comparable observation before treatment with the first comparable observation after treatment. Only these field-observed pairs can train effectiveness by disease, mixture, dose, block and weather; reconstructed weather pressure is context, not proof of treatment success.

AI and learning supervision

Admin -> AI is the full monitoring page for learning processes. It shows model version, data-through date, evidence counts, represented seasons, validation method, measured accuracy or error, data-quality findings and release gates. A model can be active while remaining review-gated. Low accuracy, too few seasons, missing Agronomist decisions or missing paired outcomes must be shown as limitations rather than converted into confidence.

Current learning processes include laboratory vintage projection, harvest-date learning, Agronomist treatment-pattern learning, disease-pressure calibration, disease onset, treatment effectiveness, product duration and retreatment cadence, FRAC rotation, young-vine nutrition, block disease calibration, spray-window learning and automated data-quality detection. New treatment, scouting, laboratory and Agronomist-review evidence rebuilds the connected learning records and feeds the live Agronomy, Treatments, Laboratory, Harvest, Alerts, WhatsApp and Admin views. Human approval boundaries remain authoritative everywhere.

Harvest prediction pipeline

```
Historical harvest dates and GDD model
  + current weather and ensemble spread
  + phenology, scouting and maturity samples
  + reviewed grape laboratory chemistry
  + damage-chain approved/provisional loss
  + treatment PHI, work readiness and cellar capacity
  -> variety forecast date/window and yield range
  -> confidence, evidence age and missing-input warnings
  -> authorized harvest plan
  -> actual pick record becomes the historical fact
```

The learned model retrains when authoritative evidence changes and reports represented years, training count, back-test error, confidence, and important factors. On-site weather and verified field/laboratory evidence carry the operational weight. Open-Meteo ensemble spread expresses near-term uncertainty; SIAS validates regional conditions; Sentinel-2 supplies block-level vegetation trends; ECMWF seasonal anomalies support early planning only. None of the external sources can independently order a pick.

Laboratory ingestion and enology pipeline

```
Original report or image
  -> retain viewable source
  -> detect every sample section on every page
  -> extract sample/lot, Annata, analytes, values and units
  -> split multi-sample reports into linked result groups
  -> assign grape variety, wine lot and vintage
  -> compare source/result signature for duplicates
  -> human review of conflicts and assignments
  -> save once -> refresh harvest or enology intelligence
```

`Annata` normally means the wine vintage. A report date in the following calendar year does not move the result to the newer vintage. For grape samples, variety and sample date route the result to harvest readiness. For must and wine samples, the linked lot and vintage route it to enology and cellar history. Exact source/result signatures prevent a report or extracted result from being imported twice.

Messaging and document-ingestion pipeline

```
Gmail / WhatsApp / Apple message / uploaded document
  -> store source and sender context
  -> classify likely domain and extract explicit facts
  -> show uncertainty, conflicts and proposed records
  -> reviewer approves, corrects or rejects
  -> approved record enters its domain pipeline
  -> replies or notifications require an explicit send action
```

If no explicit facts can be supported, the item stays in review instead of generating a guessed record. Messages cannot complete treatments, clear reminders, approve payments, alter harvest dates, or create financial facts merely because their wording resembles an action.

Hospitality inquiry, reservation and partner-commission pipeline

```
Matched inbound subject or manual inquiry
  -> guest inquiry and response history
  -> reservation conversion and availability check
  -> optional partner assignment and commission rule
  -> estimated commission while the visit is tentative
  -> earned/due commission when the qualifying visit is confirmed
  -> approval -> one or more partner payments
  -> paid/void state and Finance payable reconciliation
```

Partners are managed as hospitality business records with contact, tax/payment notes, default commission method, rate, and active state. Each reservation keeps its own commission snapshot so later partner-rate changes do not rewrite history. Estimated commissions are not Finance payables. Only earned due, approved, or partially paid balances enter the payable view; cancelled or disqualified reservations void the commission while preserving the audit trail.

Alert and operational-action pipeline

```
Sensor, scheduler, inventory or database condition
  -> evaluate threshold and evidence freshness
  -> create/update one durable alert
  -> rank by severity and route configured copies
  -> show the same authoritative state on web and TV
  -> clear only when the measured condition or owner action resolves
```

Alerts are durable database records. Delivery channels send copies; they do not own alert state. A reminder, work-plan item, or notification cannot be marked satisfied by an unrelated duplicate or stale completion.

Decision authority summary

Pipeline	System may calculate	Final authority
Damage	Provisional loss, range, confidence, trend	Agronomist-approved chain final
Treatments	Pressure, products, mixture, stock gap, date window	Agronomist approval and completed application record
Harvest	Variety date/window, yield range, confidence	Authorized plan; actual pick is final fact
Laboratory	Extraction, sample grouping, vintage proposal, duplicate check	Reviewed saved result and linked lot/vintage
Messaging	Classification and proposed records	Human review and explicit send/save action
Hospitality	Availability, lifecycle prompts, commission calculation	Authorized reservation and commission/payment actions
Alerts	Detection, severity, routing and automatic clear evidence	Authoritative measured state or explicit resolution

OPERATING REFERENCE

7. Vineyard, harvest, and historical records

Harvest records

The system stores planned quantities separately from actual harvest lots. Actual pick dates, weights, crates, blocks, varieties, and source evidence drive historical reports and forecasting.

A single picking lot can include multiple legal cadastral parcels. Select every contributing parcel when recording the pick; the database keeps each parcel as a separate authoritative relationship rather than combining the references into notes. When that picking lot is transferred into a cellar lot and tank, every linked parcel follows the trace automatically. Additional picking lots can enter the same tank and their fruit and must quantities accumulate.

Each live and printed tank label lists the deduplicated legal provenance for its current wine lot: municipality, cadastral sheet, parcel number, vineyard area, tenure, and contract protocol when recorded. The label derives this list from the retained harvest-to-tank trace and does not require the parcel information to be typed again.

Authoritative harvest dates currently include:

Vintage	Grecanico	Grenache	Nerello
2023	23 Sep	24 Sep	8 Oct
2024	11 Sep	23 Sep	23 Sep
2025	11 Sep	17 Sep	23 Sep

The 2024 Grenache and Nerello harvest date is shared. Earlier conflicting date lists were discarded by owner instruction.

Year switching

Changing the year must change the visible harvest, work, laboratory, treatment, weather, olive, labor, and financial context. Current live data and historical evidence use the same database, so comparisons do not depend on reopening old workbooks.

OPERATING REFERENCE

8. Harvest prediction logic

Harvest predictions are decision support, not autonomous picking instructions. Sebastian or another authorized person confirms the operational decision.

Authoritative inputs

- Confirmed historical pick dates by variety.
- On-site GW2000 weather history.
- Base-10 growing degree days (GDD).
- Current deterministic weather forecast.
- Fresh ensemble forecast spread.
- Variety-assigned grape laboratory results.
- Maturity samples, phenology, and field reports.
- Treatment clearance and pre-harvest intervals.
- Work readiness and cellar capacity.
- Historical model error and year-over-year behavior.

GDD calculation

Daily GDD uses:

```
max(0, ((daily minimum C + daily maximum C) / 2) - 10)
```

If minimum and maximum are unavailable, an approved daily-mean fallback may be used and is labeled. The calculation accumulates from the configured season start and is compared with variety-specific historical behavior.

Learned-model behavior

The learned model uses prior vintages as training evidence. It reports training sample count, represented years, confidence, and back-tested mean absolute error. It is recalculated when authoritative evidence changes.

New grape laboratory results, maturity samples, phenology observations, harvest plans, treatment changes, or harvest records create a durable refresh request. The scheduler processes the request rather than waiting for the normal full prediction interval.

Laboratory influence

Fresh, reviewed, variety-assigned grape laboratory data can affect readiness. Unreviewed reports or grape results older than 21 days cannot directly move the date. As harvest approaches, laboratory chemistry and field maturity receive more weight than coarse seasonal outlooks.

External sources and limits

Source	Use	Restriction
Open-Meteo ensemble	Near-term uncertainty and forecast spread	Automatic date influence bounded to plus/minus one day
SIAS regional data	Independent validation	Does not replace the on-site station
Sentinel-2	Parcel/block vegetation trend evidence (NDVI, NDRE, LAI estimate)	Cannot independently select a picking date
ECMWF seasonal	Early rainfall and temperature planning	Coarse scale; cannot set an exact date

The system uses free data without required credentials for these sources. Sentinel processing currently recognizes seven verified cadastral parcel polygons and reports the geometry scope explicitly.

OPERATING REFERENCE

9. Treatments and safety

The treatment workflow is **plan -> review -> approve -> apply -> record**.

- A forecast or model recommendation is not an approved treatment.
- A planned treatment is not a completed application.
- Completion requires actual date, product, rate, area, operator, and safety evidence when applicable.
- Agronomist review remains required for disease and treatment recommendations.
- Pre-harvest and re-entry intervals must be preserved.
- Missing safety detail is visibly flagged rather than guessed.

Projection configuration

The Home Assistant add-on configuration provides two planning defaults:

- `treatment_planning_water_l`: the provisional total carrier-water volume. It defaults to 400 L and can be changed in the Treatments page for a live calculation.
- `treatment_default_sprayer`: the name, model, or internal ID of the preferred active spray-equipment profile. The Treatments page also provides a selector when more than one profile exists.

Sprayer capacity, usable fill, nozzle setup, flow, pressure, travel speed, carrier rate and calibration status remain database equipment records. A nominal tank size may split a projection into batches, but the proposal remains blocked until the actual usable fill and field calibration are verified. Product projection requires a verified formulation, estate authorization, current crop/target use, documented rate and compatible units. Water-based and per-hectare rates are calculated directly; density or mass/volume conversions are never guessed. Products with incomplete directions remain visibly marked as needing configuration and are not automatically included.

Current source-backed product directions include these explicit method limits:

- Frontiere: vineyard foliar support at 0.75–1.00 L/ha; it is not automatic disease control.
- Ferticus 18 M: vineyard foliar nutrition at 300–500 g/100 L. The system preserves this as a mass rate and never displays it as millilitres.
- TerraPlus Solub NPK 8-7-6: 15–30 kg/ha per pass as a separate, localized soil-directed spray (or fertigation). It must not be presented as a canopy spray, must not be mixed automatically with crop-protection products, and its mother solution must not be acidified or mixed with calcium.
- Ossiclor 20 Blu Flow: vineyard copper application at 1.7–4.2 L/ha under the current authorized use, with crop stage, annual copper limits, PHI, weather, PPE and Agronomist approval checked before use.

Support and nutrition products remain unselected by default. A risk score alone never adds them to a treatment. Every projected use continues to require the current container directions and human approval.

The current audit has no open treatment safety-gap failures. Seven older applications remain explicitly restricted historical evidence because contemporaneous label, calibration or exact-use facts cannot be reconstructed. Restricted history may inform chronology but cannot be reused as verified safety evidence.

Fertilizer and young-vine nutrition separation

Fertilizer procurement shows land and whole-vineyard soil products only. Vine foliar products do not belong in that procurement list. TerraPlus is reserved for the mapped section of small, young vines and appears in the young-vine nutrition evidence instead of the general fertilizer recommendation. The system may recommend TerraPlus only when mapped vine age plus recorded weak growth, chlorosis, establishment stress, verified deficiency, tissue evidence or soil evidence supports review. It never recommends TerraPlus simply because it was purchased or is on hand.

OPERATING REFERENCE

10. Laboratory reports

The laboratory section stores original reports, sample identity, vintage assignment, matrix, variety or lot linkage, analytes, values, units, review state, and interpretation.

Correct operating sequence

1. Import or upload the original report.
2. Confirm sample name, sample type, report date, and vintage.
3. Link grape reports to a variety and wine reports to the correct lot when possible.
4. Check units and decimal interpretation.
5. Review extracted results.
6. Save the review decision.
7. Allow the prediction or cellar decision-support refresh to run.

The current finding card analyzes only the newest measured report for the selected vintage and states its source and review boundary. The measured trajectory and projected endpoint compare only the same normalized wine identity, sample type, process stage, analyte and unit. Historical endpoints use the final matching measured result from each earlier vintage. AI-assisted values are recalculated when a new numeric result arrives and are constrained to physically possible nonnegative values. An approved marker is displayed separately from a prediction and is never invented when a variety standard has not been configured.

Historical names are normalized for comparison while the original source label is retained. `Nerello`, `Nerello Mascalese`, `Narello Macalase` and vintage-suffixed forms refer to the same Nerello Mascalese identity. Documented Grecanico/Bianco-Grecanico and Grenache spelling variants are handled similarly. Grape, must, wine and other sample types remain separate and are not merged merely because their names match.

The durable laboratory model stores each prediction at its historical cutoff and scores it only against a later actual result. Future measurements never enter an earlier prediction input. Direction accuracy below the release threshold keeps projections review-gated. Nine laboratory records are currently flagged for genuine source review; the authoritative report index has no missing, incomplete, wrong-type or duplicate sample groups after normalization.

OPERATING REFERENCE

11. Cellar and tank labels

The cellar module tracks physical containers, wine lots, volume, phase, manual or sensor readings, and operations. A planned container assignment is not the same as physical occupancy.

Tank labels

Each physical tank keeps a stable label URL. Changing the assigned wine updates the label contents without changing the tank URL. Labels include:

- Tank identity and correct vessel type.
- Capacity, contained volume, and calculated level.
- Wine/vintage/lot information.
- Recent temperature, density, Brix, and pH readings.
- Small recent-reading charts when space permits.
- Legal company identity and cellar contact.
- Browser favicon, Apple touch icon, and Android/PWA identity.
- Responsive layouts for different tablet sizes and print layouts.

Cellar displays are enrolled inside the VPN. They can then operate through the approved outside connector. The direct Nabu Casa address is the preferred outside label route; the URL remains available as a backup when QR provisioning is unsuccessful.

OPERATING REFERENCE

12. Olive oil records and cost logic

The olive section compares harvested olive weight, oil liters, conversion efficiency, total modeled cost, and cost per liter by year.

Confirmed 2024 result

- Olives: 332 kg
- Oil: 40 L
- Conversion: 8.3 kg olives per liter
- Oil yield: 12.048 percent by the system's liters-per-kilogram comparison

Editable 2024 cost assumptions

- Pressing: EUR 0.20 per kg of olives.

- Bottle and label: EUR 2.30 per 500 ml bottle.
- Planned bottles: 220.
- Supplier invoice: EUR 751 net plus 22 percent VAT.
- Annual labor: EUR 1,000.
- Harvest labor reference: EUR 540 at EUR 7 per tree, included in annual labor by default.

The supplier invoice is treated as including pressing and bottling by default, preventing duplicate cost counting. The correct formulas produce:

- Pressing component: 332 x EUR 0.20 = EUR 66.40.
- Bottling component: 220 x EUR 2.30 = EUR 506.00.
- Supplier gross: EUR 751 x 1.22 = EUR 916.22.
- Total with annual labor: EUR 916.22 + EUR 1,000 = EUR 1,916.22.
- Cost per actual oil liter: EUR 47.91.
- Actual 500 ml bottle equivalents from 40 L: 80.

The planned 220 bottles require 110 L, so they exceed the confirmed 2024 oil volume by 70 L. Cost assumptions can be edited and saved per year.

Olive harvest outlook

The Olive page displays an estimated harvest date and uncertainty window from exact Baiamonte olive harvest dates stored in MariaDB. Year-only notes are excluded. With only a small number of exact seasons, the system correctly reports low confidence. A current-year recorded harvest date replaces the estimate and is labeled as an actual date. The calendar estimate is planning support only; fruit maturity, oil accumulation, crop condition, weather, and mill availability still require field confirmation.

Separate olive treatment program

Every treatment belongs to either **Vineyard** or **Olives**. The two histories, forecasts, summaries, and entry controls are separate. Vineyard disease-pressure scores are never reused to predict an olive treatment. Olive forecasting remains unavailable until olive-specific scouting or trap observations identify a target.

The recorded 2026 olive treatment is retained with its exact products, dose basis, water volume, and calculated totals. Historical 2025 workbook entries are retained as completed olive work, but their missing products, doses, target, and safety details remain visibly unverified.

The system can suggest what to apply only after a database record confirms the exact crop, target, current Italian authorization, recent label verification, dose range, PHI, REI, and related restrictions. If those fields are incomplete, the product recommendation fails closed. The Agronomist must approve every candidate before application.

OPERATING REFERENCE

13. Labor, contractor invoices, and payments

Labor and contractor records move through review and payment states. Hours, fixed services, reimbursements, and invoice payments are stored separately enough to preserve an audit trail.

Standard workflow

1. Worker or manager records hours, service, expense, or invoice evidence.

2. The record is submitted for review.
3. An administrator corrects or approves it.
4. Approval locks the payable basis.
5. One or more deposits or payments can be recorded against an invoice.
6. The payment ledger determines paid, part-paid, or unpaid status.
7. Fully paid items do not reappear in the payment queue.

The current payment-integrity audit shows no paid-ledger mismatches, no fully paid items reappearing, no partial-payment status errors, and no verification holds. Six non-payable records are marked paid; they are tracked as a data-quality category and do not create a current payment mismatch. The formerly future-dated labor entry was corrected and is no longer an open source-review exception.

OPERATING REFERENCE

14. Finance

Fatture in Cloud is mirrored read-only. Vineyard Operations does not write back to the accounting provider. Finance access is limited to authorized finance users and is excluded from TV, kiosk, and public feeds.

Use the Finance section to review documents, parties, VAT context, balances, payment status, linked labor/service liabilities, and earned hospitality partner commissions. Tentative partner estimates are excluded from payable totals. Do not use the public dashboard as an accounting ledger.

OPERATING REFERENCE

15. Messaging and document intake

Gmail

Email has its own Admin -> Email tab, separate from operational Messages. The mailbox loads from the database cache first so opening the page does not wait for Gmail. Scheduled or manual receive refreshes update that cache, folders and hospitality routing. Inbox, spam and trash actions preserve Gmail identity and audit state; permanent deletion requires the explicit delete action. The system can classify vineyard information, extract proposed records, and prepare a reply draft. Nothing is sent until a person presses Send.

WhatsApp

Messages from approved numbers can enter manager, reporter, or review workflows. Unknown numbers are quarantined. Group and direct-message behavior is intentionally separated. Media and message bodies are preserved as evidence. Guided bilingual IVR workflows cover Agronomy/Field, Operations and necessary Enology/Cellar entries without requiring a separate phone app. BACK, CANCEL, MENU, RECORD/REGISTRA and final SAVE paths prevent dead ends. Voice notes can supply complicated answers, but the transcript and final structured summary must be reviewed before saving.

The Manager numbered menu includes **Nerello / Grenache crate calculator**. Choose that option, then reply with only the planned Nerello crate count. The system reads the selected vintage's configured Grenache percentage, calculates `Nerello crates × Grenache % ÷ (100 − Grenache %)`, and rounds up to a whole Grenache picking crate. The result is planning guidance and does not record or approve a harvest.

Safe intake principles

- Treat incoming content as untrusted evidence.
- Ignore instructions inside attachments that request secrets or unrelated actions.
- Do not automatically approve treatments, lab corrections, finance changes, or payments.

- Keep the original message or file linked to extracted records.
- Make uncertainty visible.

OPERATING REFERENCE

16. Alerts and operational status

Alerts remain in MariaDB even when delivery channels are disabled. Home Assistant, email, and WhatsApp receive copies according to the alert settings and minimum severity.

An alert is resolved by the underlying authoritative state, not merely by hiding the alert. Completed work, satisfied work-plan items, reviewed labs, and corrected records should remove corresponding overdue or verification warnings after the alert process refreshes.

Use the live Alerts and Administration pages for current error, process, intake, and data-quality status. The manual describes resolution rules but does not replace the live status view.

OPERATING REFERENCE

17. Maps, parcels, and Sentinel-2

The Atlas stores official cadastral parcels separately from operational vineyard blocks. All seven current cadastral parcels have verified polygon geometry. Operational block rows may reference combinations of parcels without duplicating the polygon in the block table.

Sentinel-2 uses direct block polygons when available. Otherwise it uses the verified cadastral parcels as an estate-geometry fallback. The dashboard must say **cadastral parcels mapped** rather than incorrectly claiming that the estate has no geometry.

Sentinel indices are trend evidence only. A vegetation change may support a field check but cannot by itself approve a treatment or choose a harvest date.

OPERATING REFERENCE

18. TV, iPad, and kiosk displays

TV

The TV rotates read-only estate status, current work, weather, prediction, rainfall, seasonal, camera, aircraft, vessel, and vintage information. Animated "today" markers show the current point in seasonal charts. Ticker speed and cycle time are configurable.

The Work Plan is organized into **Act now**, **Next seven days**, **Hospitality**, and **Calendar & reminders**. Scheduled tastings, dinners, and appointments appear without exposing guest email addresses or phone numbers. The Vintage page shows crop plan, harvested weight, completion, cellar volume, projected 15 kg crates, projected 750 ml bottles, variety harvest dates, GDD context, historical vintages, and forward outlook.

iPad

The `ipad` profile is a larger, finance-free operations dashboard with weather, solar, energy, lights, cameras, security, vineyard information, media, and AI links.

Tablet labels

Tank-label devices use a dedicated enrollment flow. Each registered tablet has a permanent label URL and its own QR code. Already visited label pages, branding, and the last successful tank reading are cached so a label can reopen and remain useful during a temporary connection outage.

For a factory-reset Android tablet, open **Enology -> Tablet setup** and scan the managed-device QR during Android's initial setup. The add-on hosts a checksum-pinned Fully Kiosk Browser EMM installer locally and supplies a private one-page profile that launches the assigned label after boot, stays in landscape kiosk mode, recovers after connectivity changes, and enables local-network Remote Admin. The separate Start URL QR remains available as a manual fallback. Reprovision controls should replace the device identity cleanly rather than creating duplicate tablets.

OPERATING REFERENCE

19. Hospitality

Hospitality is an internal, low-volume booking and service workspace for private estate experiences. The current release supports one private guest party at a time and is designed for tastings and dinners for approximately 6 to 12 guests.

Packages

Administrators and Hospitality Managers can maintain the active package name, description, minimum and maximum guest count, duration, base price, per-person price, deposit requirement, inclusions, and preparation notes. The initial packages are:

- Private Estate Tasting: 1 to 6 guests.
- Cellar Tasting & Pairing: 2 to 8 guests.
- Private Estate Dinner: 6 to 12 guests.

Pricing remains configurable and quote-based. Saving a package changes future selection options; it does not silently rewrite a confirmed guest quote.

Gmail inquiry routing

Hospitality Admin controls a list of inbound Gmail subject phrases. The initial phrase is **Inquiry about Reserve Tasting**. Matching is case-insensitive and ignores common Reply and Forward prefixes. More phrases may be added one per line without changing protected Gmail credentials.

Matched public messages enter **Hospitality -> Guest inquiries** even when the sender is not on the vineyard operations allowlist. They are safe requests, not approved operational records. Attachments continue through the protected intake safeguards. Existing matching Gmail records are reconciled as well as new mail.

Open an inquiry to read it, prepare and explicitly send an email response, change its state, add internal notes, or convert it into a reservation. Conversion pre-fills the available guest identity and source, then requires the operator to choose the package, date, time, guest count, and booking state. The inquiry is linked to the resulting reservation.

Reservation workflow

1. Record the inquiry and select a package, date, time, and guest count.
2. Capture the guest name and only the contact details needed for service.
3. Record dietary restrictions, accessibility needs, celebration details, preferences, internal notes, quote, and deposit.
4. Move the reservation through requested, confirmed, arrived, completed, cancelled, declined, or no-show states.
5. The server rejects overlapping confirmed or arrived experiences so only one private guest party is committed at a time.

6. Send email or WhatsApp confirmation only by pressing the explicit communication action. Phone calls and notes can be recorded without sending a message.
7. Open any reservation to update its details or delete it after confirmation. Deletions remain in the audit trail; a linked inquiry returns to the responded queue.

The hospitality communication log preserves channel, subject or summary, operator, delivery state, and timestamp. Guest contact details remain inside the protected workspace and are intentionally omitted from TV data.

Partner management and commissions

Hospitality Admin includes a partner database for concierges, hotels, travel advisors, event organizers, and other referral sources. A partner record stores the organization and contact details, commission method and default rate, tax/payment notes, status, and internal operating notes.

Assigning a partner to a reservation creates a reservation-specific commission snapshot. A percentage rule uses the eligible reservation amount; a fixed rule uses the saved fixed amount. The operator may change the reservation commission without changing the partner's future default.

Commission states follow the reservation lifecycle:

1. Inquiry or requested reservation: estimated only; not payable in Finance.
2. Qualifying confirmed/arrived/completed reservation: earned and due.
3. Approved commission: authorized for payment.
4. Partial payment: remaining balance stays due.
5. Paid: payment ledger equals the earned amount.
6. Cancelled, declined, no-show, or otherwise disqualified reservation: void, with history retained.

One commission can receive multiple payments. Each payment preserves amount, date, method, reference, note, and operator. Corrections are audited; deleting a payment is restricted and does not erase the original commission or reservation relationship. Finance includes only earned outstanding partner balances and separates them from tentative estimates.

OPERATING REFERENCE

20. Register

The Register is a large-touch sales workspace intended for a tablet in the United States. It combines three controlled sources:

- Sellable winery products, prices, and stock mirrored read-only from Fatture in Cloud.
- Active hospitality packages maintained locally under Hospitality.
- Authorized manual items created in Register Admin.

The Register has four pages: **Sale**, **Inventory**, **Ledger**, and **Admin**. Completed sales are written to the local MariaDB ledger. Register sales are not posted to Fatture in Cloud in release 1.6; the future posting fields exist but remain locked off until a separate reconciliation release is approved.

Checkout, currency, and language

1. Tap catalog items, adjust quantity or editable price, and apply any authorized line discount.
2. Select **EUR** or **USD** directly at checkout. EUR remains the accounting and VAT reporting base.
3. Select **English** or **Italiano**. This selection controls the PayPal checkout locale and the printed receipt language.

4. Confirm the active PayPal Business account. EUR prefers the Italian account when configured; USD prefers the US account. An authorized operator may select the other configured account.
5. Complete payment by cash, hosted PayPal/card checkout, or operator-confirmed PayPal Tap to Pay.

For every transaction the database preserves the subtotal, discount, VAT, and total in EUR; the actual tender currency and amount; the exact saved USD-per-EUR rate; the selected PayPal account; the checkout language; the payment or capture reference; and the operator identity. The receipt and CSV export carry the same reconciliation fields.

The ECB rate refresh is a convenient reference rate, not a claim that it is a bank settlement rate. Administrators may save the actual checkout rate when required; the saved rate remains attached to the sale even after the current setting changes.

PayPal account setup

Protected Home Assistant App Configuration holds credentials for both accounts:

- **US PayPal Business:** the existing PayPal client ID and secret.
- **Italian PayPal Business:** the Italian PayPal client ID and secret.
- **Environment:** Sandbox for testing or Live for real payments.

The browser receives only the selected public client ID. Client secrets remain server-side. Never enter, store, or transmit raw card numbers through the Baiamonte application.

Tap to Pay and receipts

PayPal Tap to Pay runs in the PayPal POS application on an NFC-capable phone. After PayPal approves the exact amount, the operator records the PayPal transaction reference in the Register. This is explicitly an operator confirmation; the web browser does not pretend to verify the contactless charge in real time.

The system receipt can be printed through the browser's configured system printer. It is labeled as an operational, non-fiscal receipt. The monthly Ledger shows both collected tender and EUR base values and exports a UTF-8 CSV for reconciliation.

OPERATING REFERENCE

21. Scheduled processes and recovery

Process	Typical interval	Function
Complete refresh	60 min	Recovers missing or overdue subsystems
Planning	15 min	Syncs Baiamonte Calendar and Tasks
Weather	15 min	Imports on-site current and historical weather
Forecast sources	180 min	Refreshes ensemble, SIAS, Sentinel, and seasonal evidence
Harvest prediction	90 min plus queued refreshes	Recalculates harvest readiness
Gmail	15 min	Imports approved messages and attachments
WhatsApp	15 min	Refreshes connection and approved catalogs
Finance	360 min	Pulls read-only accounting data
Etna	15 min	Updates volcanic and seismic context
Disease	30 min	Updates disease and heat-stress support
Alerts	5 min	Evaluates operational warnings

The complete refresh is a recovery sweep. It does not blindly duplicate every job; it reruns subsystems that are missing or substantially overdue. Manual **Run complete update now** executes the configured subsystems immediately.

OPERATING REFERENCE

22. MCP and Codex integration

The Baiamonte MCP server is a constrained interface for Codex and approved automation.

Connection

```
[mcp_servers.baiamonte]
url = "http://192.168.0.10:8100/mcp"
bearer_token_env_var = "BAIAMONTE_MCP_TOKEN"
```

The endpoint is available locally or through the VPN. It requires the bearer token for every MCP request. Unauthenticated requests are rejected.

Tool groups

The live server exposes 23 tools:

- **Status and planning:** processing status, work plan, Apple reminder lists, Baiamonte reminders, treatment reminders.
- **Review intake:** queue review item.
- **Read models:** vineyard overview, harvest report, lab history, lab decision context, weather summary, open issues, finance overview, funding report, record search.
- **Confirmed writes:** tasks, vineyard records, lab results, lab reviews, financial documents, cash transactions, and funding-requirement updates.

MCP writes are currently enabled. Every write tool still requires explicit confirmation of the exact record. Read tools can be used for diagnosis without changing data.

Version interpretation

Home Assistant add-on version 1.6.42 is the operator-facing release. The MCP protocol handshake may report a different server-framework version; that framework identity must not be used as the Vineyard Operations update version.

OPERATING REFERENCE

23. Security and privacy

- Keep passwords, API keys, and tokens in Home Assistant add-on configuration or environment variables.
- Never place a bearer token directly in documentation, screenshots, logs, or a repository.
- Use role-based Home Assistant accounts.
- Keep finance out of shared displays.
- Keep guest contact, dietary, accessibility, and private-event details out of shared displays and public feeds.
- Keep MCP on the local network/VPN unless a separately authenticated tunnel is deliberately configured.
- Review MCP writes before enabling them on a new client.
- Use recoverable deletion for mail and records where supported.
- Preserve original evidence and audit events.

MCP credential rule

The Baiamonte MCP connection uses `BAIAMONTE_MCP_TOKEN`. Tokens belong in a protected environment variable or connector-secret mechanism, never inside a URL, screenshot, ticket, or documentation file. Rotate a token immediately if it is exposed.

OPERATING REFERENCE

24. Troubleshooting

A page has no data

1. Confirm the selected year.
2. Refresh the page once.
3. Check Administration -> Processing Log.
4. Check whether the source process is enabled and recent.
5. Verify that the record is approved and assigned to the correct year/vintage.
6. Do not enter replacement zeroes for unknown data.

A map looks stale

Refresh the page after geometry changes and reopen Atlas so the map recalculates its visible size. If parcels still appear outside Sicily or the geometry count is wrong, check the Atlas process and saved cadastral geometry before editing a block.

An alert remains after work is done

Confirm the authoritative work record is completed, then run or wait for the Alerts process. Hiding a card does not resolve the underlying alert.

A payment reappears

Check its payment ledger, invoice total, deposits, payment timestamps, and approval state. The current integrity audit reports no fully paid invoices reappearing.

Hospitality does not open

Confirm the Home Assistant username is linked to an Administrator or Hospitality access profile. A title alone does not grant access unless the profile is saved. Administrators configured in protected add-on settings are also recognized before a matching People profile is created.

Register or a PayPal account does not open

Confirm the Home Assistant username has Administrator, Register, Cashier, or Hospitality Manager access. In Register Admin, the active connection line identifies Fatture in Cloud and each configured PayPal account. If an account shows **not configured**, add that account's client ID and secret in protected App Configuration and restart the add-on. Use Sandbox until both the amount and currency have been verified end to end.

A hospitality event is missing from the TV

Confirm the reservation is requested, confirmed, or arrived and has a future or recent start time. Cancelled, declined, completed, and no-show reservations remain in history but do not occupy the active TV schedule.

MCP does not connect

1. Confirm VPN/local reachability to port 8100.

2. Confirm `BAIAMONTE_MCP_TOKEN` is present in the client environment.
3. Confirm the Home Assistant MCP bearer token is configured.
4. Confirm the client host is permitted.
5. Restart Codex after configuration changes.
6. Use `processing_status` as the first read-only test.

OPERATING REFERENCE

25. Release 1.6.42 operational snapshot

Release additions

- The official Ministry catalog is written in small transactional batches so retaining the complete source record does not exceed the live MariaDB per-query timeout.
- Agronomy now imports the official Italian Ministry plant-protection catalog weekly or on demand and overlays registration, authorization status, expiry, active substances, and formulation evidence on current Baiamonte products.
- Exact trusted matches that are revoked, suspended, expired, or absent from the latest official catalog are blocked from treatment projections.
- Products outside the estate's historical use pattern can enter the simulator only after the Agronomist records the current authorized crop, target, dose limits, PHI, REI, and label source.
- First-use products remain low-confidence, require explicit Agronomist approval, and create a paired pre/post-treatment scouting requirement so estate-specific outcomes can accumulate.
- The owner manual now includes the complete authoritative release history from 1.6.0 through 1.6.40, generated directly from the project changelog so maintenance fixes and smaller workflow changes remain part of the durable operator record.
- The downloadable PDF and in-window manual preview expand to 37 verified pages with synchronized release metadata and page navigation.
- Agronomy and Enology cold loads wait until every feature renderer is registered, and changing vintage preserves the current workspace and page.
- The laboratory source audit normalizes documented aliases, reports no missing authoritative samples, and prevents negative endpoint projections.
- Disease and treatment intelligence now exposes weather rationale, onset, Agronomist calibration, paired scouting outcomes, block calibration, spray-window learning, duration, cadence and resistance rotation under explicit evidence gates.
- The treatment simulator follows the Baiamonte one-pass process using two 200 L fills and includes only currently justified products.
- Admin -> AI monitors every connected model, validation measure, evidence count, data-quality finding and human-review boundary.
- Gmail opens from a database-backed mailbox cache in its own Admin -> Email tab, while guided WhatsApp and voice workflows cover field, operations and essential cellar entry without dead ends.
- The manual now documents the complete intelligence architecture, its evidence router, authority boundaries, and decision trees for damage, treatments, harvest, laboratory ingestion, messaging, hospitality commissions, and alerts.
- Damage-chain cards preserve each report's own AI result while the event summary retains the latest proposal, agronomist-approved final, and forecast-effective loss as separate values.
- Hospitality partner records, reservation-linked commission rules, due/approval/payment states, multi-payment history, and Finance payable reconciliation are available without treating tentative estimates as debt.
- A dedicated Register workspace is available between Hospitality and Admin and is optimized for tablet operation.
- Sellable wine inventory mirrors read-only Fatture in Cloud products; hospitality products remain local; manual sellable items are supported.

- Checkout supports EUR and USD while retaining EUR as the authoritative reporting base.
- Both the Italian and US PayPal Business accounts can be configured, visibly selected, and retained on each sale.
- English and Italian checkout/receipt language can be switched before payment.
- Hosted PayPal/card checkout and operator-confirmed PayPal Tap to Pay preserve capture or POS references without storing raw card data.
- Receipts and monthly CSV exports preserve EUR base, collected tender, exchange rate, account, language, and payment reference.
- Register and Cashier are distinct access roles managed through Administrator People.
- A dedicated Hospitality workspace is available beside Operations and Admin.
- Hospitality Manager is a distinct role and access profile.
- Home Assistant identity and Vineyard Operations authorization are synchronized without creating duplicate people records.
- Three configurable experience packages and the complete reservation lifecycle are installed.
- Server-side conflict control enforces the one-private-party operating model.
- Hospitality confirmations are explicit-send actions with an audit history.
- The TV Work Plan includes scheduled hospitality and has a clearer four-panel layout.
- The TV Vintage page includes crop, harvest, cellar, package-output, schedule, GDD, history, and outlook context.
- The administrator authorization path and the Hospitality endpoint now use the same access rules.
- Configurable Gmail subjects route public requests into a dedicated Guest inquiries queue.
- Inquiries support explicit email response, status changes, notes, deletion, and reservation conversion.
- Reservations support audited updates and deletion, and their dialogs scroll safely on short screens.
- Hospitality now has Reservations, Guest inquiries, and Admin tabs; package and Gmail routing controls are grouped under Hospitality Admin.
- Hospitality navigation now uses the same shared menu behavior as Operations and Admin on desktop and mobile, and restores the last selected Hospitality section.
- Payroll Control moved from documentation into Operations Control.
- Treatment completion is authoritative only when recorded on the treatment itself. Checking off a Google or Apple reminder cannot mark an application as completed, and an open current reminder is not cleared by a stale completed copy from another source.
- Treatment 5 is completed on 27 June 2026 following the estate-wide hailstorm on the evening of 26 June. The owner-confirmed standard vineyard volume is 400 L; the six product rates and calculated totals are authoritative, while the exact time, operator, treated scope, weather and safety confirmations remain pending.
- Olive and vineyard treatments are separate programs. The 2026 olive treatment and vineyard treatments 2-4 now use owner-supplied source sheets; the two 2025 olive treatments retain their exact workbook row provenance.
- Olive harvest timing has its own confidence-labeled historical calendar model and does not share the grape harvest model.
- Old overdue plans remain visible until completed, cancelled, or rescheduled. Unconfirmed completions do not affect vineyard treatment recency.
- Product prediction fails closed until current crop-and-target authorization and label evidence are stored.

- Digital tank labels use larger Baiamonte branding, clearer typography, and the same physical container silhouettes and stage-driven liquid motion as the cellar TV.
- Tank-label shells and the last successful readings remain available offline, while enrollment and revoked links remain uncached.
- Android kiosk provisioning uses a locally hosted, checksum-verified Fully Kiosk EMM installer with managed autostart, recovery, landscape, kiosk, and local Remote Admin settings.
- The enology workspace groups the most-used controls compactly and keeps long history and AI review sections expandable.
- Today weather shows the current condition, sensor freshness, wind and gusts, soil moisture, UV and active vineyard advice; the Weather page adds pressure, solar radiation, dew point, VPD, forecast and severe-weather actions.
- Weather history and all operational calculations read MariaDB records. Home Assistant remains a live sensor and forecast input, not a separate authority.
- Weather-file upload and every workbook migration command entry point are retired. Historical workbook provenance may remain attached to migrated database records, but no workbook is read by the running system.
- Condition alerts resolve automatically when their measured or service condition clears and matching Home Assistant notifications are dismissed.
- Olive harvest strategy is an owner-selectable database setting. The greener-harvest forecast is calculated from recorded evidence and current conditions rather than a fixed September date.
- The operating-history chart begins with the first actual estate vintage in 2023; the invalid pre-operation 2022 grape harvest is rejected during migration.
- Today uses the same severity-ordered database alert feed for its count, urgent finding and complete scrolling notice list. Historical vintage review is labelled separately from live estate systems.
- Etna, trends, projections, finance summaries and historical records are database-authoritative. Workbook files and workbook import commands are not runtime authorities.
- Digital labels use the current transparent Baiamonte mark and the same physical, liquid-confined vessel animation as the cellar television display.

Verification completed for this release

- The application and MariaDB health check passed after installation.
- Administrator Hospitality access passed; an unassigned operations account was correctly denied.
- Seed packages, database migrations, television feed, grape rows, forecast structure, and cellar tanks were verified on the running installation.
- The complete automated suite passed before publication, including Today presentation, olive forecasting, Etna/trends authority, alert lifecycle and database-authority safeguards.

Source-review items remain visible rather than being guessed. Laboratory assignments, treatment safety details, inventory receipts, and planned container sharing stay in their respective review queues until authoritative evidence resolves them.

OPERATING REFERENCE

Appendix A. Complete release coverage: 1.6.0-1.6.42

The operational chapters above describe the cumulative current system. For completeness, the following appendix includes every published change recorded for the full 1.6 release series, from 1.6.0 through the current 1.6.42 release. It is generated directly from the authoritative project changelog when the manual is built, so maintenance fixes and smaller workflow changes are not omitted from the owner record.

Release 1.6.42

- Uses small transactional Ministry-catalog write batches so the full official JSON audit record imports within the live Home Assistant MariaDB per-query timeout.
- Reports zero rather than an unknown count before the first successful official-catalog synchronization.

Release 1.6.41

- Adds the official Italian Ministry plant-protection product catalog as a weekly, manually refreshable regulatory source.
- Overlays exact registration and reviewable name matches on Baiamonte products; trusted revoked, suspended or expired products are blocked from projected treatments.
- Allows an Agronomist to add a product outside the estate's historical use pattern only after recording its current crop, target, dose limits, PHI, REI and authorized label source.
- Marks products without completed Baiamonte applications as low-confidence first-use candidates and requires Agronomist approval plus paired pre/post-treatment scouting.
- Adds an Agronomy treatment panel for official catalog search, local overlay review, product adoption and crop-use authorization without treating national registration as a label direction.

Release 1.6.40

- Expands the owner system manual with the complete authoritative release history from 1.6.0 through 1.6.40.
- Generates the release appendix directly from the project changelog so maintenance fixes and smaller workflow changes are retained in the durable operator record.
- Rebuilds the downloadable PDF and all 36 in-window preview pages with synchronized release metadata and navigation.

Release 1.6.39

- Updates the full owner manual through the current Agronomy, Enology, Laboratory, Admin AI, Gmail-cache and guided WhatsApp workflows.
- Documents the weather-first disease model, Agronomist review, paired scouting, learning gates, one-pass two-batch treatment process and young-vine TerraPlus boundary.
- Rebuilds the downloadable PDF and every in-window manual preview page with current release metadata.

Release 1.6.38

- Starts the dashboard only after every Agronomy and Enology feature renderer is registered, eliminating intermittent cold-load failures.
- Preserves the active Agronomy or Enology workspace immediately while changing vintages instead of temporarily resetting navigation to Operations.

- Normalizes authoritative laboratory aliases and vintage suffixes during source auditing and corrects four source-confirmed historical sample types.
- Constrains laboratory endpoint and AI-assisted projections to physically possible nonnegative values while retaining the measured adjustment as evidence.
- Guards the embedded weather-map observer against an unavailable document root during iframe startup.

Release 1.6.37

- Corrects the worker portal task queue to use the current planned/in-progress statuses and normal priority, restoring assigned work visibility.
- Makes guided WhatsApp issue priorities compatible with the Operations schema and adds the missing lot/tank reference to cellar-operation submissions.
- Expires abandoned WhatsApp forms and calculators after 24 hours, and makes MENU reliably close either workflow before returning to the main profile menu.
- Updates FastAPI, Starlette, the multipart parser and MCP runtime to secure compatible releases after dependency auditing; the resulting environment reports no known vulnerabilities.

Release 1.6.36

- Replaces the partial WhatsApp intake menu with complete bilingual guided workflows for Agronomy/Field, Operations and Enology/Cellar records.
- Adds saveable field scouting, phenology, planned treatment reports, completed work, labor hours, issues/tasks, equipment service, fruit maturity, fermentation checks and cellar operations.
- Makes WhatsApp voice notes first-class form answers; complicated reports can be spoken once, reviewed as a transcript summary and explicitly saved into the Operations review queue.
- Adds consistent BACK, CANCEL, MENU, RECORD/REGISTRA and final SAVE recovery paths so active sessions do not strand workers or silently create records.
- Keeps treatment field reports planned and unapproved until the Agronomist completes the legal, safety, weather and product review.
- Adds Admin WhatsApp IVR health, active/stalled session counts, 24-hour completions, voice readiness, workflow coverage and worker command guidance.

Release 1.6.35

- Adds durable disease-onset forecasts that project when weather-driven pressure may cross the actionable threshold, with walk-forward validation and explicit confidence.
- Learns treatment effectiveness, observed duration and retreatment cadence from paired pre/post scouting by disease, mixture, dose, block and weather without treating reconstructed pressure as proof of causation.
- Adds chronological active-ingredient and FRAC-group rotation review, blocking repeated groups for Agronomist review when the current product records support the comparison.
- Adds young-vine TerraPlus learning from mapped applications, growth scouting, soil and tissue evidence while retaining the documented-need and Agronomist-approval gates.
- Adds adaptive sensor, duplicate-observation and unreliable-label detection to operational data quality and Admin → AI.
- Adds bounded block-specific disease calibration from parcel/map context, canopy/training system, elevation/aspect, variety and localized scouting.
- Adds outcome-conditioned spray-window learning from wind, temperature, carrier coverage and rain in the first 48 hours after application; legal label and Agronomist limits remain authoritative.

- Rebuilds all connected learning manifests after treatment, scouting and Agronomist review, exposes them in Admin → AI, and provides an administrator rebuild endpoint.

Release 1.6.34

- Adds a durable disease-pressure calibration model trained only from explicit Agronomist decisions and comparable field scouting while retaining the weather/phenology rules score as an auditable baseline.
- Applies bounded, small-sample-shrunk disease adjustments to current pressure, Treatments, alerts, WhatsApp and AI consumers; sparse feedback can never move a score by more than 20 points.
- Adds corrected risk scores to the Agronomist review workflow and automatically rebuilds disease learning after reviews, scouting, startup and scheduled pressure refreshes.
- Adds leave-one-case-out calibration error, training-label provenance, season coverage and validation gates to Admin → AI; validation requires at least eight labels across two seasons and performance no worse than the rules baseline.

Release 1.6.33

- Adds an audited Agronomist review workflow to current disease-pressure assessments; rejected findings stop driving treatment predictions, while materially changed scores return to pending review.
- Adds treatment-linked before/after scouting with strict block, date-window and disease-marker checks, plus conservative automatic pairing only when one treatment match is unambiguous.
- Rebuilds treatment outcome learning and Admin → AI evidence metrics immediately after paired scouting, while retaining weather as context rather than proof of effectiveness.
- Adds a Treatments follow-up panel for missing baselines, due or overdue post-treatment checks, completed pairs and observed outcomes.
- Caches Gmail folders and message summaries in MariaDB during scheduled intake so Email opens from local data; explicit receive and refresh controls still perform live Gmail checks.

Release 1.6.32

- Adds a dedicated **Admin** → **AI** console containing OpenAI service and credit status, effort and speed controls, budget and projected cost, feature usage, model pricing, and durable-learning health.
- Monitors Laboratory, Agronomist treatment, Harvest, and disease-pressure intelligence with explicit model versions, evidence coverage, data freshness, validation methods, accuracy or error metrics, readiness thresholds, and review-gate warnings.
- Distinguishes measured held-out accuracy from deterministic rules: disease pressure reports Agronomist review coverage without claiming outcome-trained accuracy, and an unavailable model cannot hide the other learning processes.
- Keeps Operations Control focused on system operations and limits the Audit Trail viewport to the latest six visible activities while retaining the latest 100 entries in a scrollable list.

Release 1.6.31

- Keeps the durable Laboratory model in an explicit learning-active, low-accuracy state when historical walk-forward direction accuracy is below 60%, even when case and vintage coverage thresholds are met.
- Runs both Laboratory and Treatment durable-learning backfills during application startup so newly installed migrations and historical evidence are reflected immediately in their current pipelines.

Release 1.6.30

- Adds durable vineyard-treatment learning with pre-treatment weather, leaf wetness, solar, phenology, scouting, objectives and treatment cadence; 14-day post-treatment evidence windows are cut off before the next application.
- Separates weather-reconstructed pressure from field-observed effectiveness, versions the treatment model and feature schema, calibrates similarity from the 2023–2026 GW2000 history, and keeps the model provisional until adequate seasons and comparable scouting outcomes exist.
- Normalizes historical laboratory identities while retaining every original report label; Nerello, Narello and Nerello/Narello Macalase resolve to the canonical Nerello Mascalese series, alongside documented Grecanico and Grenache aliases.
- Adds durable laboratory cutoff cases, leakage-safe historical walk-forward outcomes, versioned model manifests, per-analyte-and-unit error metrics, and automatic rebuilding at startup and after every new, corrected or reviewed laboratory result.
- Exposes transparent Laboratory and Treatment learning status in their current pipelines and APIs without converting predictions into automatic agronomy, cellar, harvest or enology approvals.

Release 1.6.29

- Passes the latest GW2000 disease-pressure snapshot into current treatment simulations so weather-pattern similarity is calculated rather than left unavailable.
- Learns the authoritative product/date/water program from every structured completed treatment, including safety-restricted Treatment 5, while continuing to prohibit historical safety evidence from authorizing a current prescription.

Release 1.6.28

- Narrows TERRAPLUS SOLUB NPK 8-7-6 to the small, young vines only and labels its procurement card accordingly.
- Removes TERRAPLUS from general mature-vine, olive and whole-estate nutrition baselines; it remains unavailable unless mapped young vines have a documented need and an Agronomist-directed root-zone or fertigation plan.

Release 1.6.27

- Corrects TERRAPLUS SOLUB NPK 8-7-6 from broadcast land/soil fertilizer to vineyard root nutrition for fertigation or localized root-zone application.
- Separates land/soil fertilizer procurement from vine root-nutrition procurement so TERRAPLUS appears with vine nutrition without being mislabeled as a foliar canopy spray.

Release 1.6.26

- Rebuilds vineyard treatment simulation around the Agronomist's complete Treatments 2–5 rather than selecting isolated products, while retaining current label, necessity, exact-mixture, weather, PHI/REI, PPE, inventory and Agronomist approval gates.
- Calculates every weather-matched historical program for the current Baiamonte process: one vineyard pass using 400 L total carrier prepared as two identical 200 L batches.
- Makes GW2000 weather the explicit treatment rationale by comparing temperature, humidity, 72-hour and seven-day rain, soil moisture and other available markers with the weather preceding completed treatments.
- Adds continuous, leakage-safe learning: each confirmed completed vineyard treatment stores only the seven days of weather available before application, reconstructed disease pressure, the full recipe and a stable program signature.

- Uses the learned weather match in both live next-treatment review predictions and the simulator, with transparent similarity, model-version, training-case and leave-one-treatment-out validation evidence.
- Keeps restricted historical treatments as behavioral learning evidence only; they never become current reusable prescriptions without fresh safety and authorization review.

Release 1.6.25

- Rebuilds the Agronomy treatment simulator around the Baiamonte field process: one complete vineyard pass using 400 L total carrier prepared as two identical 200 L batches.
- Leads with a readable full-treatment recipe showing each necessary product, the amount in each 200 L batch, the whole-treatment amount, inventory readiness, and the current evidence supporting its inclusion.
- Prevents inventory and prior use alone from promoting support, adjuvant, or nutrition products; those products require documented current stress or another explicit evidence-supported need.
- Screens independently significant same-date disease pressure for additional necessary control while retaining exact-mixture compatibility, label, weather, safety, and Agronomist approval gates.
- Adds Laboratory variety-standard cards that show recorded approved analyte markers by variety, sample type, stage, and unit, while clearly leaving missing standards unconfigured.

Release 1.6.24

- Removes the red Live cellar records readiness banner from the Cellar page; real tank alerts remain visible, while the informational Demo-mode banner is retained only when Demo mode is active.
- Preserves user-selected values systemwide when live refreshes rebuild a select box's options, including the Laboratory Sample and Measurement selectors, without restoring values that are no longer valid choices.

Release 1.6.23

- Preserves the former shared Live/Sandbox choice on the first upgrade to independent US and Italian PayPal selectors, even when Home Assistant pre-populates the new fields with schema defaults.
- Records the one-time migration in persistent add-on data so later operator changes to either account remain independent and are never overwritten on restart.

Release 1.6.22

- Adds independent Live/Sandbox selectors for the US and Italian PayPal Business accounts in protected Home Assistant configuration.
- Uses the selected environment for each account's OAuth token, order creation, capture, checkout label, and status light, preventing one account's mode from changing the other.
- Keeps the former shared PayPal environment as a migration fallback for existing installations.
- Makes guest-inquiry deletion persistent by retaining an invisible tombstone linked to the Gmail intake item, so dashboard refreshes and later Gmail checks cannot recreate a deleted inquiry.
- Keeps Labels & dedicated displays selected after saving a tablet assignment instead of returning to the Cellar overview.
- Adds live Online/Offline heartbeat lights to each manually registered cellar tablet, including Tablet 1 and Tablet 2.
- Keeps Laboratory sample and measurement selectors useful when the selected vintage has no numeric report by clearly showing the latest available vintage, while refusing to imply that historical evidence is a current-year measurement or projection.

Release 1.6.21

- Exports the protected US and Italian PayPal client IDs, client secrets, and environment from Home Assistant app options into the Register API process.
- Restores the already-configured US PayPal Business account in checkout while preserving the existing EUR-to-US-account fallback when no Italian account is configured.
- Preserves the explicitly selected PayPal environment from protected Home Assistant options; the Baiamonte installation is configured for live PayPal after deployment.
- Adds persistent US and Italian PayPal Business status lights to the sale page, showing Active/Offline and LIVE/SANDBOX before checkout.
- Compacts the oversized Register header pill to a short FIC, PayPal-account, and environment summary; the detailed sync timestamp remains in Integration status.
- Moves Register category buttons to a dedicated wrapping row so Wine, Oil, Hospitality, and later categories are never clipped by the search field or Manual item button.

Release 1.6.20

- Routes clear guest tasting, visit, tour, dinner, and reservation inquiries into Hospitality even when the Gmail label or configured subject phrase is missing; the messages remain review-only until explicitly converted.
- Rechecks previously downloaded Gmail message bodies so the existing “Inquiry about Classic Tasting at the Estate” request can enter Guest inquiries.
- Restores saved Atlas parcel and block shapes with an SVG overlay that remains correctly sized when Agronomy is opened after page startup.
- Separates saved boundary counts from parcel center locations so Atlas no longer reports point-only records as verified boundaries.
- Keeps the active workspace, page, and subpage selected when changing years instead of returning the navigation to Operations.
- Suppresses cellar monitor alerts for tanks whose recorded volume is zero; readings remain visible, and guard checks resume automatically when a positive volume is recorded.
- Makes every Finance summary metric clickable and keyboard accessible, opening the selected year’s supporting invoices, monthly result calculation, cash accounts, or individual approved labor and payment records.

Release 1.6.19

- Added transparent AI-assisted Laboratory projections: exact prior-vintage evidence is preferred, with a low-confidence 14-day current measured trajectory only when at least two dated readings exist; every refresh and new lab arrival recalculates the outlook.
- Added newest-report AI-assisted finding cards to Laboratory and Vineyard fertilization, with source/review limitations and no automatic cellar, harvest, product, rate, or application decision.
- Restricted Fertilizer procurement to explicitly classified land/soil products; foliar vine-treatment fertilizers remain in Treatments and Crop nutrition.
- Added a dedicated Email tab beside Messaging in Administration, containing mailbox folders, reading/actions, attachments, received/sent history, connection controls, intake refresh, and compose.
- Splits the oversized Laboratory series menu into Sample and Measurement selectors.
- Keeps samples with comparable prior-vintage evidence first while retaining clearly labeled current-only samples.
- Limits the measurement menu to the selected sample, sample type, and stage so unlike records cannot be chosen accidentally.

- Makes “Mark Enologist approved” save immediately and clears the sample’s source-review flag so approved reports stop appearing red.

Release 1.6.18

- Opens Laboratory on a series that actually has comparable prior-vintage evidence instead of defaulting to a current-only measurement.
- Groups the selector into measurements with comparable history and measurements that have only a current reading, so unavailable projections are explicit.
- Replaces the misleading mixed-sample annual-average chart with like-for-like vintage endpoints for the selected wine, sample type, stage, analyte, and unit.
- Replaces the process-wide mixed table with the final measured endpoint for that same sample definition in each available vintage.
- Refreshes report history, decision status, trends, comparisons, and projections immediately after a laboratory result is corrected, then reopens the detail with the saved value.

Release 1.6.17

- Consolidates safe laboratory label variants such as Grecanico 25/2025, Bianco–Grecanico, Granache/Grenache, and Nerello/Nerello Mascalese for vintage comparison.
- Keeps unrelated wines, sample types, process stages, analytes, and units separate so projections do not combine unlike measurements.
- Adds sample type and process stage to the Laboratory selector, removing visually duplicated choices and making grape, must, and wine series explicit.
- Adds Giancarlo's source-backed December 2024–November 2025 monthly labor to reconciliation: 988 known hours plus one 15-day attendance record whose hours remain explicitly unknown.
- Applies the owner-confirmed €10/hour rate to known historical hours, records all prior labor as paid with ledger entries for the 11 known-value months, and preserves November 2025 as paid with its unknown hours and amount explicit.
- Prevents mirrored source rows from double-counting labor totals and excludes paid historical attendance with an unknown amount from the zero-value invoice warning.

Release 1.6.16

- Keeps the active Enology, Laboratory, Bottling, Fertilization, or Nutrition page selected while changing vintages.
- Orders vintage-specific refreshes after the shared year state changes and ignores stale responses from rapid year switching.
- Repairs the Laboratory outlook endpoint under Home Assistant ingress so recorded vintage measurements and projections load instead of an empty fallback.
- Reloads analyte comparisons on every vintage change and exposes the existing source-backed 2023–2025 bottle-equivalent backfill in Bottling.

Release 1.6.15

- Defers direct-route activation until the signed-in workspace permissions are available, restoring Laboratory and Messaging links.
- Safely binds the later-loaded laboratory outlook renderer so opening Labs cannot interrupt the page.
- Preserves the Tank Details workspace-save repair and verified Etna status from the previous patches.

Release 1.6.14

- Completes the workspace startup repair by safely binding the later-loaded cellar-history renderer.
- Restores the remaining Agronomy, Etna, Laboratory, and Messaging initialization path exposed during live 1.6.13 verification.

Release 1.6.13

- Repairs Admin Messaging when Review has already loaded the shared communications data.
- Rebuilds the Messaging panel placement and restores its Contacts, Meta Business, and System-account tab state whenever the page opens.
- Prevents the later-loaded analytics bundle from stopping dashboard startup before Agronomy/Etna, Laboratory navigation, and Messaging are initialized.
- Keeps Tank Details and other Agronomy saves in the current workspace and page instead of returning to Today.
- Keeps direct navigation and cached communications responsive without requiring a full dashboard reload.
- Adds regression coverage for Messaging page activation and stale-panel recovery.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.12

- Isolates Messaging, Social, and TV browser startup so an error in one page cannot prevent the other pages from loading.
- Adds an independent page-loader fallback for direct navigation and restored sessions.
- Keeps hospitality Gmail inquiry refreshes active while restoring access to the shared Messaging interface.
- Adds regression coverage for page startup isolation and hospitality inquiry synchronization.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.11

- Aligns the Admin Docs card with the installed owner’s manual release 1.6.4 instead of the obsolete 1.5.0 label.
- Updates regression coverage so the displayed manual release cannot silently drift again.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.10

- Routes Gmail messages tagged with the configured Hospitality label into Guest Inquiries, including nested labels and messages that were downloaded before the label was applied.
- Retains subject-based hospitality routing as a fallback and exposes configurable inbound labels and subjects in Hospitality Admin.
- Adds a manual Gmail refresh action and regression coverage for label normalization, re-routing, and source metadata.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.9

- Records harvest picks from multiple vineyard blocks while preserving every legal parcel in the grape-to-tank traceability chain.

- Treats the entered crate weight as net kilograms per crate, calculates the field total, and supports a later authoritative winery weight without erasing the original field measurement.
- Keeps the complete harvest form visible on phones and separates harvest routes from the main application module for easier maintenance.
- Restores laboratory outlooks when switching years by using each report's authoritative vintage and keeps current-vintage projections separate from historical comparisons.
- Resolves Today greetings through the linked Home Assistant person or worker identity while excluding tablet, kiosk, iPad, and service accounts.
- Adds product images to register items and a fast editable manual-charge item while preserving local ledger and inventory behavior.
- Adds regression coverage for harvest weighing and multiple blocks, parcel traceability, laboratory year selection, register behavior, and greeting privacy.
- Leaves the full-version manual unchanged because this is an operational patch release.

Release 1.6.7

- Prevents overlapping System Documentation refreshes from leaving the control stuck on “Loading…”.
- Adds a Rome-time greeting on Today and includes a first name only when Home Assistant supplies a natural full human name.
- Shows continuously updated application uptime in Admin Control.
- Adds live Up/Down heartbeat badges for active tank-label tablets based on their latest check-in.
- Adds regression coverage for greeting privacy, documentation refresh recovery, uptime, and label-device status.
- Leaves the full-version manual unchanged because this is an operational reliability patch release.

Release 1.6.6

- Keeps every TV card and both sides of split layouts visible on narrower or overscanned displays.
- Replaces unsupported relational CSS selectors with explicit compatibility classes for older Samsung and embedded TV browsers.
- Loads vessel rendering before the main TV bundle and reduces continuous chart and overflow work for smoother long-running displays.
- Adds regression coverage for TV layout preservation, legacy-browser CSS, helper order, and refresh throttling.
- Leaves the full-version manual unchanged because this is a TV reliability patch release.

Release 1.6.5

- Reuses healthy MariaDB connections through a bounded, thread-safe pool instead of reconnecting for every small dashboard query.
- Compresses large API and TV-display responses for substantially lower tablet and kiosk network transfer.
- Preserves the existing short-lived TV payload cache while making cold refreshes and concurrent displays more efficient.
- Leaves the full-version manual unchanged because this is a performance patch release.

Release 1.6.4

- Adds a complete hospitality partner directory with referral attribution, configurable commission rules, due and approval states, partial payments, payment history, and auditable corrections.
- Connects partner liabilities to the Finance payable total and provides a dedicated partner payment queue and annual summary.
- Expands the System Manual to 24 portrait pages with intelligence-pipeline descriptions, decision trees, evidence gates, partner workflows, and current operational guidance.
- Rebuilds the downloadable PDF and in-window page previews for the updated manual.

Release 1.6.3

- Adds Home Assistant-managed network receipt printing through a configured `notify`, `script`, or `shell_command` service.
- Lets administrators select browser printing or the Home Assistant IP-printer path and optionally provide a target printer entity.
- Preserves the existing browser receipt as an automatic fallback when the Home Assistant printer service is unavailable.
- Keeps receipt print counts and audit history consistent for both printer paths.
- Leaves the full-version manual unchanged for this patch release.

Release 1.6.2

- Adds an explicit touch-friendly cash-received confirmation with amount due, cash tendered, and change retained in the audit trail.
- Allows administrators to correct payment metadata or void a local payment while preserving its audit history and automatically restoring unposted inventory.
- Protects captured online PayPal payments from local deletion and directs refunds through PayPal before ledger reconciliation.
- Replaces register payment browser prompts with purpose-built tablet dialogs, clarifies the operator-confirmed PayPal phone workflow, and keeps online PayPal API verification separate.
- Refreshes the ECB EUR/USD reference rate once daily while keeping the checkout rate editable and preserving the applied rate on every sale.
- Compacts register item cards and administration controls, and standardizes register buttons for clearer touch targets and states.
- Leaves the full-version manual unchanged for this patch release.

Release 1.6.1

- Corrects Giancarlo Pafumi's paid €20 gas reimbursement from the erroneous future date 27 August 2026 to the source-supported date 27 July 2026.
- Repairs the archived July timesheet source alongside the labor ledger and records both before/after values in the audit log without changing the amount or paid status.
- Corrects the archived source title from Mattia to Giancarlo and leaves the full-version manual unchanged.

Release 1.6.0

- Adds a tablet-oriented Register workspace with Sale, Inventory, Ledger, and Admin pages plus dedicated Register and Cashier access.
- Mirrors sellable Fatture in Cloud products and prices read-only, includes local Hospitality packages, and supports authorized manual items, editable prices, and discounts.

- Keeps EUR as the authoritative reporting and VAT base while allowing EUR or USD collection at the saved checkout conversion rate.
- Supports both Italian and US PayPal Business credentials, shows the active account at checkout, and preserves the selected account on each transaction.
- Adds an English/Italian checkout switch that controls PayPal locale and the printed receipt language.
- Supports hosted PayPal/card checkout and operator-confirmed PayPal Tap to Pay on an NFC phone without storing raw card data or claiming browser-side POS verification.
- Preserves EUR base values, collected currency and amount, conversion rate, PayPal account, language, and payment reference in receipts and monthly CSV exports.
- Keeps register sales posting to Fatture in Cloud disabled until a separately approved reconciliation release.
- Updates and visually verifies the portrait, scrollable System Manual for the full 1.6 release.

OPERATING REFERENCE

26. Glossary

Actual: A completed, confirmed event or measurement. **Evidence:** The source message, document, sensor, note, or record supporting a value. **GDD:** Growing degree days, a temperature accumulation measure. **Ingress:** Home Assistant's authenticated route into an add-on interface. **Intelligence proposal:** A traceable calculated result that remains separate from an approved fact.

Pipeline: The ordered evidence, calculation, review, approval, and refresh path for a domain decision.

Provisional: A system result temporarily available for planning while its required human review is pending. **Hospitality Manager:** The role responsible for guest inquiries, packages, private experiences, guest readiness, deposits, and confirmations.

Cashier / Register user: A limited role that can operate estate sales without receiving finance, vineyard-write, payroll, or administration access.

EUR base: The authoritative euro value used for VAT and management reporting even when the customer pays in USD.

Tender amount: The amount actually collected in the selected payment currency. **MCP:** Model Context Protocol, the controlled interface used by Codex and approved automation. **NDVI / NDRE:** Satellite vegetation indices used as trend evidence. **Planned:** Intended but not completed or approved. **Review item:** A visible uncertainty that must not be guessed. **Vintage:** The harvest year to which grapes or wine belong, even when later work occurs in another calendar year. **YoY:** Year-over-year comparison.

OPERATING REFERENCE

27. Operating principle

The system is designed to be useful without pretending to know more than the evidence supports. Confirmed facts remain authoritative, forecasts remain provisional, unknowns remain visible, and sensitive actions remain under human control.